

Natural Language Processing with Large Language Models

Exercises Spring Term 2026

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DS-NLP Lab

Exercises: Time to work on your Foundation Model



~15 min *Theory in Code (notebook)*

- Transformer Architecture
- Attention

Co-Working & Mentoring Time

Sync with your co-founders and build your product.

- Ask questions & get individual support
- Work **onsite** with your group



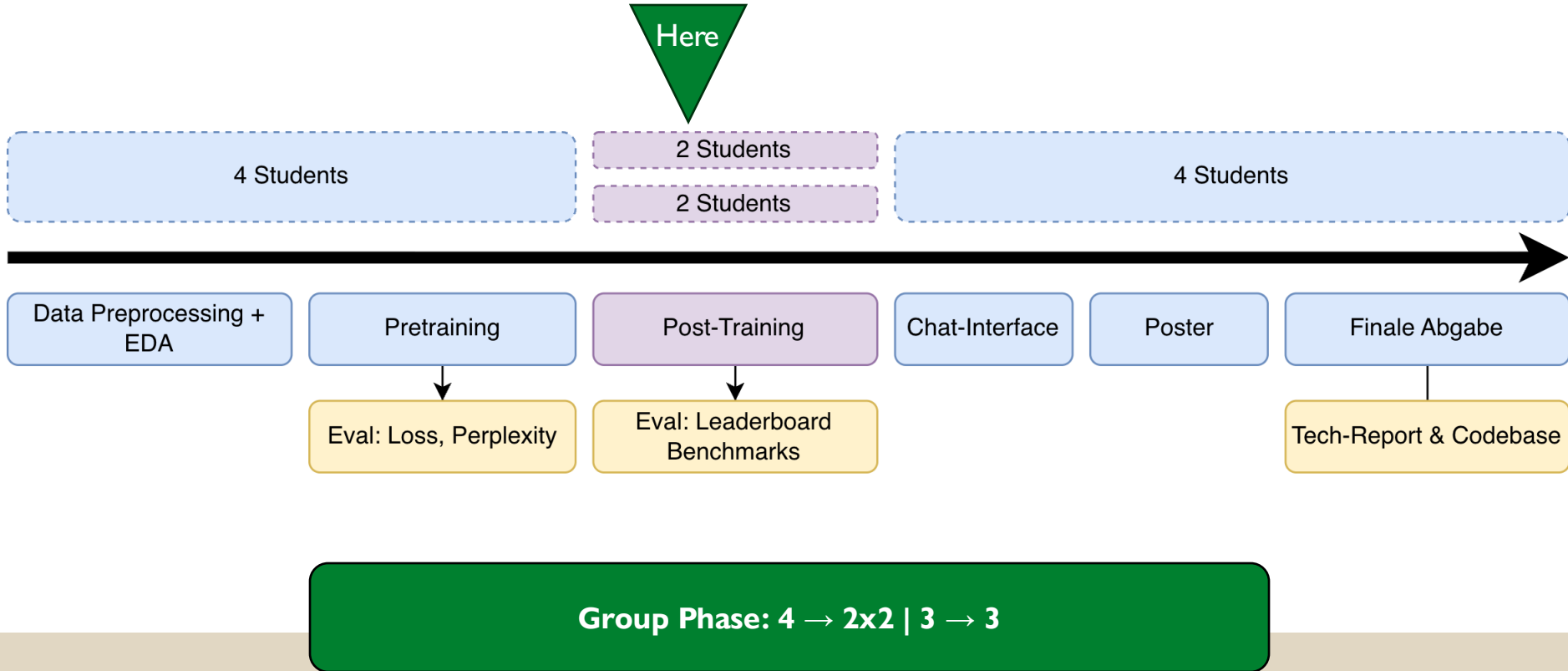
Reminder: Dates and planning

	13.04			Slide submission via Canvas
7	14.04	Vadian Gallus - EG		Midterm Presentation
8	21.04	Vadian Gallus - EG	20. - 05.05. Lorena: ICLR 2026 Rio de Janeiro	
9	28.04	Bodensee - 1.OG		
10	05.05	C, 61 61-070		
	11.05			Poster submission via Canvas
11	12.05	Hilti Inn. R2 - 1.OG		
12	19.05	Hilti Inn. R1 - 1.OG	Pseudo-Conference	Demo & Poster
	02.06		Project Submission	Code + Report Submission



Lorena is not available from 20-05.05.

Roadmap: Group Phases

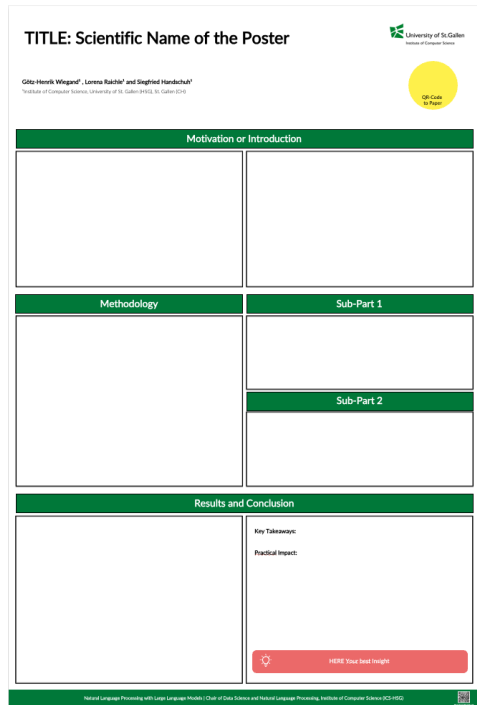
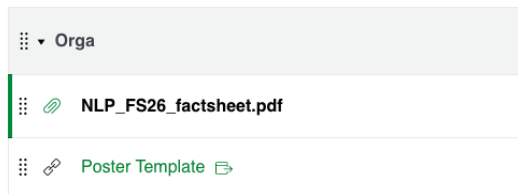


Exercise Today

- Poster Template
- PikoGPT Leaderboard
- Theory in Code: RLHF and DPO

Poster Template

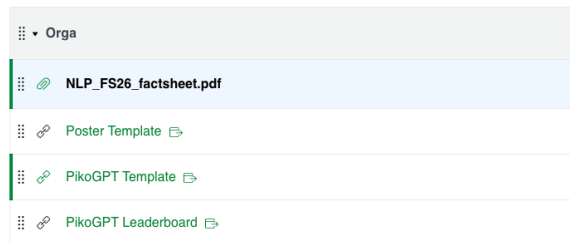
- Template provided in canvas
- One poster per startup
- How to poster:
 - Make clear visuals carry the story: "A picture tells more than 1k words."
 - Include clear scientific flow
 - Problem > Method > Results > Conclusion
 - Keep it short and simple: Focuss on the most important things!
 - Have a good 🧠-Catcher!



If you have any questions or want feedback, talk to me!

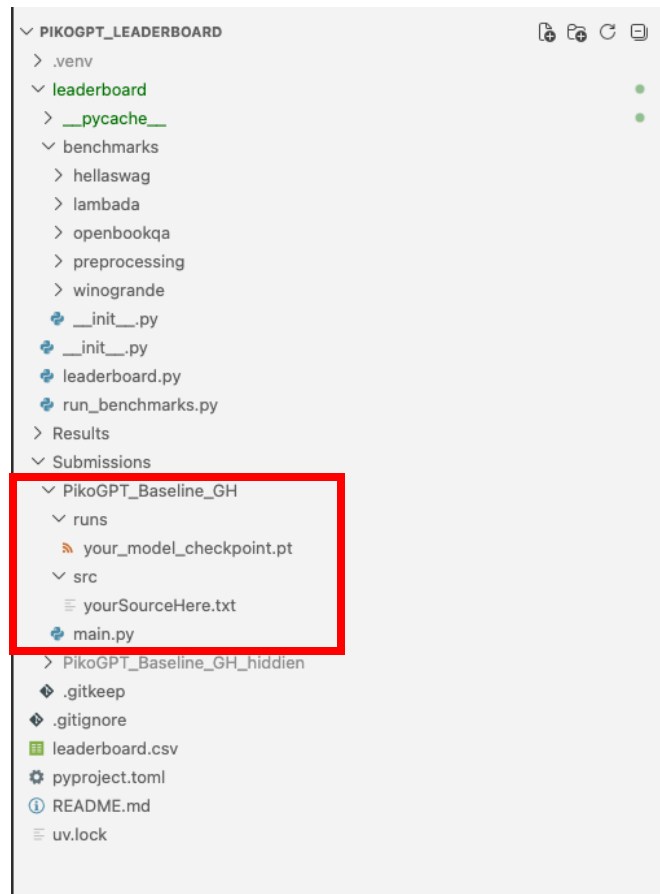
PikoGPT Leaderboard

- We have a leaderboard script -> https://github.com/unisg-ics-dsnlp/PikoGPT_Leaderboard
- How to create a Submission:
 1. Fork the Repository
 2. Create a new Folder with src-folder, main.py and model checkpoint in the submissions-folder
 3. Run the leaderboard evaluation
 4. Create a pull request (PR) with your submission.
- GH will go through the PR and give you the results for the hidden benchmarks.
- We have provided a baseline model. We expect you to be better than this.



1. How to Leaderboard Submission - Preparation

- After cloning and creating a new branch with:
 - `git switch -c YOUR_SUBMISSION_NAME`
- Create Submission
 1. Create a new folder in the Submissionn
 2. Put the main.py inside the folder
 3. Put src with everything that is needed for the inference-stage in the folder
 - See factsheet for inference-stage infos.
 4. Put your model checkpoint in for example „runs“
- Next step: run the leaderboard

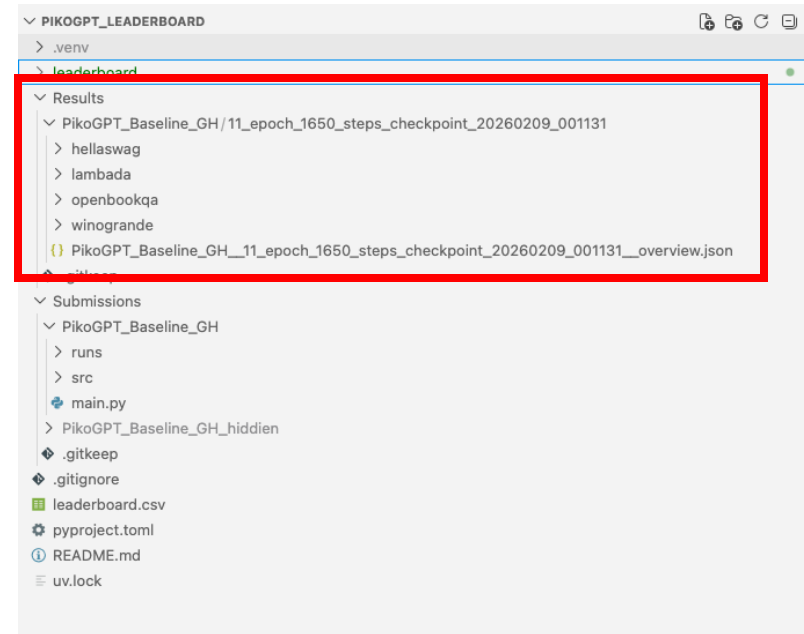


2. How to Leaderboard Submission - Run Leaderboard and Submitt

- Bevor running the full submission -> Test with small example

```
uv run python -m leaderboard.run_benchmarks \  
  --submission MyTeam \  
  --checkpoint runs/my_checkpoint.pt \  
  --limit 5
```

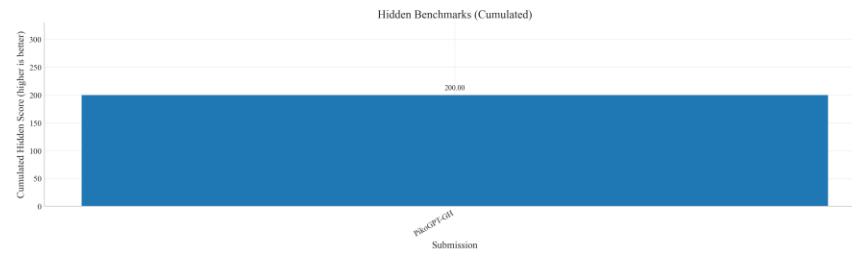
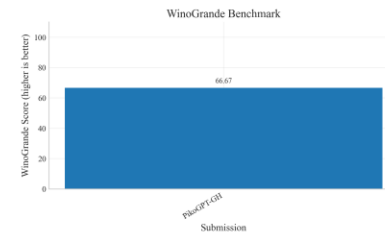
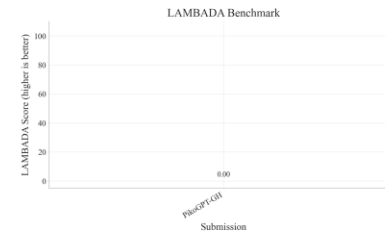
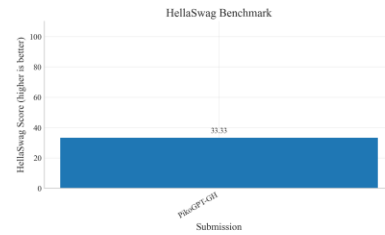
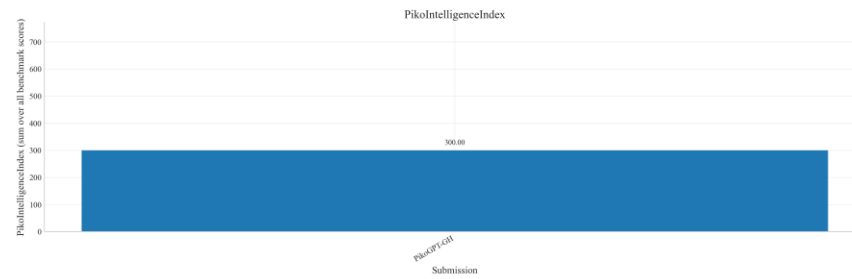
- If that works, run benchmark with limit 100
 - See README.MD for reference
- After benchmarks finished -> results in the results folder
- Now commit and push your submission to your fork on GitHub
- Final Step:
 - Create a Pull Request



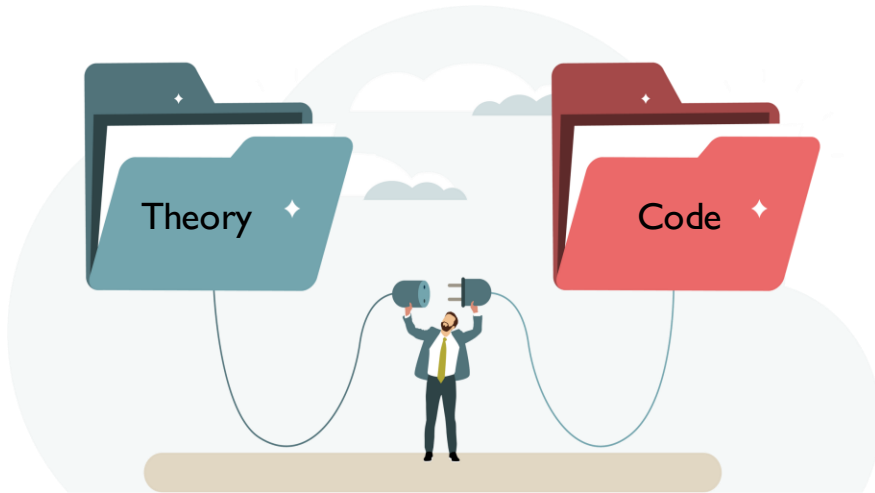
How do we Score?

- Benchmark Metrics
 - hellaswag, openbookqa, winogrande, lambada
 - One score per metric
- Hidden Benchmarks
 - We dont tell you -> hidden
 - Metric: Comulated score over all benchmarks
- Piko-Intelligence-Index
 - Cumulated Score over all Benchmarks
- Baseline Score PikoGPT_GH:
 - HellaSwag: 33.33
 - WinoGrande: 66.67
 - Lambada and OpenBookQA: 0
 - Hidden Benchmarks: 200
 - **Piko-Intelligence-Index: 300**

Important: The Hidden-Benchmarks might change over Time.



Theory in Code: Hyperparameter Tuning



Direct Preference Optimization